

Assignment 3

Solutions will be discussed during the f2f tutorials on Feb 3 and Feb 4. Submission is not required.

1. Consider a multiple regression model, $y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + u$, with the Gauss-Markov assumptions.

(a) Given the sample, write down the $Var(\hat{\beta}_j)$, where $j = 1, 2, \dots, k$. Explain every notation in detail.

(b) Explain how the three factors affect the variance.

(c) What is the difference between $sd(\hat{\beta}_j)$ and $se(\hat{\beta}_j)$.

(d) What is the consequence of having two highly correlated variables, x_1 and x_2 ?

2. Comment: “Consistency and Unbiasedness are two different concepts; one does not imply the other.”